Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 1st part–January 24th 2025

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and ensure to provide a brief explanation. Remember, without a reason, the answer will be considered invalid. Understanding the questions is crucial for your success in this exam.

A. In a PRAM model, "all processors can access all memory cells in unit time. Could you please provide a detailed explanation of the implications of this assumption? (1)

B. Could you please elaborate on the conditions under which the following formula holds as an equality and when it holds as an inequality: $SU_p \leq p$?"

C. Could you please elaborate on what "weak scaling" means? To which law we have seen during the lectures is referring to? (1)

D. Please describe the speed-up when a fixed-size model is considered. (1)

Exercise n. 2

Answer the following questions about advanced algorithms algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Could you please describe how the Strassen algorithm improves matrix multiplication? Additionally, which class of algorithms does the Strassen approach belong to? (2)

B. "Why can BDD be considered an example of a data structure that exploits dynamic programming? How does asymptotic complexity change with memoization?" (2)

Exercise n. 3

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. What are the key ideas of the basic Karger approach compared to previous approaches? Does it affect the worst-case complexity?" (2)

B. Describe how the search for Skip-list data structure works and discuss its complexity. (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the complexity of the partitioning algorithm used by the quick sort. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

B. Please discuss the different variants of the Select-ith algorithm. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**

C. Describe the ideas behind the primality test algorithm presented during the lectures. Which class of randomized approach does it belong to? (1)